

KALPANA DEVI T

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[Portfolio: kalpana-portfolio-pied.vercel.app](https://kalpana-portfolio-pied.vercel.app)

PROFESSIONAL SUMMARY

Python & AI Developer with hands-on production experience building RAG-based chatbots, AI-powered web analysis platforms, and Gmail automation systems. Skilled in FastAPI, LLM integration, Generative AI, and scalable backend architecture. Completed B.E. in Computer Science Engineering with a CGPA of 7.8. Passionate about deploying intelligent systems that solve real-world business problems.

WORK EXPERIENCE

Python & AI Developer | BRANDINGBEEZ, Coimbatore | 2025 – Present

- Developed and deployed a production-grade AI web analysis platform (brandingbeez.co.uk) for automated website scraping, keyword extraction, and SEO report generation using LLM integration, FastAPI, and Node.js.
- Designed and deployed a RAG-based AI chatbot (acgraphics.com) using FastAPI, OpenAI API, and FAISS with intent classification, brand-tone enforcement, enquiry detection, and real-time fallback mechanisms.
- Built a live Gmail automation system (Octopus.ai) using OAuth 2.0 Gmail API for automated unsubscribe processing, background task execution, and Swagger UI API documentation.
- Developed an AI-powered web scraper and article summarizer that stores structured data in Supabase with an HTML/CSS frontend for article display.

INTERNSHIP EXPERIENCE

Data Analytics / AI / Machine Learning Intern | NOVITECH R&D PVT. LTD | 2022–2024

- Completed three focused internships in Data Analytics, Artificial Intelligence, and Machine Learning — gaining hands-on experience in model building, feature engineering, and data pipelines.

IoT Intern | IYARKAI TECH LAB PVT. LTD, Chennai | 2023

- Worked on IoT-based embedded systems and developed sensor integration projects.

Business Analyst Intern | SAKTHI VEERA GREEN ENERGY PVT. LTD, Chennai | 2023

- Analyzed business processes for freeze drying machinery operations and delivered structured reports.

TECHNICAL SKILLS

Languages: Python, JavaScript, Node.js, HTML, CSS, React

AI & ML: Generative AI (GenAI), LLM Integration, RAG, Prompt Engineering, LangGraph, Dialogflow, NLP, Computer Vision, Deep Learning, Scikit-learn, Keras, OpenCV

Backend & APIs: FastAPI, RESTful APIs, API Integration, OAuth 2.0, Token-Based Authorization, Background Task Processing

Databases: Supabase, MongoDB, SQL, FAISS (Vector Indexing), Vector Databases, Data Modeling

Architecture: Modular Architecture, Scalable System Design, Configuration-Driven Development, Service-Oriented Architecture

Libraries: Pandas, NumPy, Scikit-learn, Keras, OpenCV, Puppeteer (JavaScript)

Tools: Git, GitHub, Docker, Postman, Swagger UI, VS Code, Headless Browser Automation

KEY PROJECTS

- License Plate Recognition System:** Built an ALPR system using OpenCV and Pytesseract OCR with Canny Edge Detection, contour filtering, and ROI extraction to recognize plates from images and video.
- Face Recognition System:** Developed a real-time face detection system using Haar Cascade Classifier in OpenCV with optimized frame-by-frame grayscale processing.
- Moving Object Detection:** Designed a real-time motion detection system using frame differencing, background subtraction, contour detection, and bounding box tracking in OpenCV.

EDUCATION

B.E. Computer Science Engineering — Jansons Institute of Technology, Coimbatore | 2022–2026 | CGPA: 7.8

HSC (2022): 83% | SSLC (2020): 93% — Scism. Mat. Hr. Sec. School, Bodinayakanur, Theni

CERTIFICATIONS

Programming Languages (C, C++, Python) | Data Analytics | Artificial Intelligence | Machine Learning

ACHIEVEMENTS

- Won 2nd prize in paper presentation at Karunya University.
- Selected in Top 100 teams at Hack4Purpose national-level idea presentation, Anna University, Chennai.